

ME1103 – Principles of Mechanics and Materials
STATICS AND FRICTION

VENUE: EA-04-21

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NOTE: Wear proper attire in the laboratory. No sandals, slippers or any non-covered shoes are allowed.

1) INTRODUCTION

Statics is the gateway into engineering mechanics, which is the application of Newtonian physics to design and analyze objects, systems, and structures with respect to motion, deformation, and failure. It is concerned with the analysis of applied force acting on a physical body/system that is in static equilibrium (at rest or moving at constant velocity) with its environment. It is the foundation to the study of strength of materials, which then relates the effects of external loads to the internal response of the body, in terms of the stresses induced in the body and deformation it experiences.

Statics is used in extensively engineering applications, especially in the design of structures and machines. Engineers use the principles of statics to calculate forces in structures like bridges, buildings, and mechanical devices, ensuring they can withstand the forces they encounter.

In summary, statics is an essential field of study in engineering that helps us understand and predict the behaviour of stationary objects and structures. Understanding statics gives us insights into the natural world, helping us understand how objects and structures, from the smallest machine to the largest skyscraper, withstand the forces they experience.

2) THEORY

2.1) Static Equilibrium of Rigid Body

Consider a rigid body that is subjected to multiple forces applied on it at different locations. Due to the principle of transmissibility, each of these external forces can be represented as a force-couple pair at any reference point, say point O in the figure. Upon these operations, the system of forces becomes an equivalent system of concurrent forces and moments acting at O , which can be added to give the resultant force and moment, as depicted in Fig. 1.

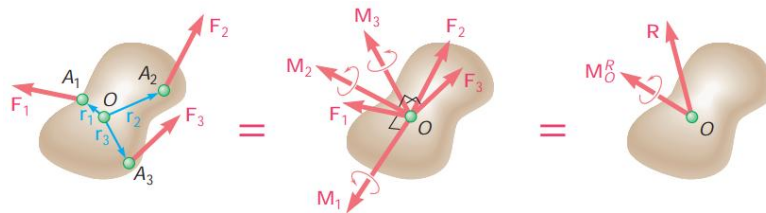


Fig. 1. Statics of a rigid body.

Finally, the rigid body is said to be in static equilibrium, when the resultant force and moment are zero, that is,

$$\vec{R} = \sum \vec{F}_i = 0 \text{ and } \vec{M}_O^R = \sum \vec{M}_i = 0 \quad (1)$$

2.2) Principle of Superposition

For a rigid body that is subjected to multiple loads, such as the simply supported beam in Fig. 2(a), where there exists a distributed load on part of the beam, and a concentrated load applied at a specific position. This problem can be analyzed by considering each load separately and then adding their individual responses together. This approach is based on the principle of superposition that states that for a linear elastic structure, the total response (reaction force or displacement) due to the multiple loads is the sum of the effects caused by each loading separately.

As such, this also implies that the response of the structure to a given load can be derived by removing the effects from the other loads, as shown in Fig. 2(b). Note that you will see the effectiveness of this principle of superposition in action when solving the mystery bar problem in section 4.1).

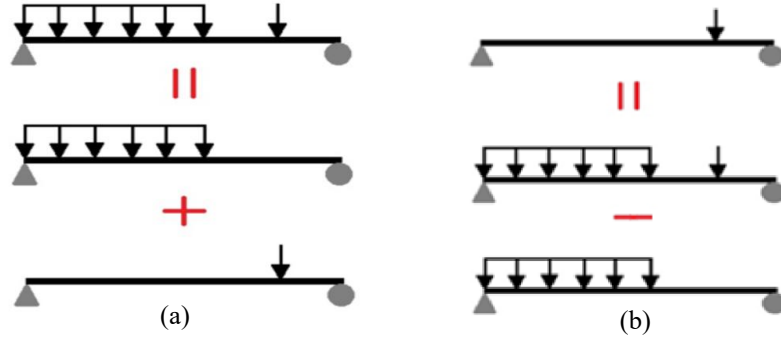


Fig. 2. Principle of Superposition on a beam.

2.3) Coefficient of Static Friction

We have a love-hate relationship with friction. Without it, we will not be able to carry out many simple tasks, such as moving around on foot or writing with pencils. On the other hand, friction is also known to be a key contributor to the failures of many mechanical systems, such as air-craft engines, due to the wear and tear that arises from the friction at the contact interfaces.

Friction force is observed experimentally to be directly proportional the normal force N at the contact interface and is dependent on the roughness of the two surfaces involved. Figure 3 shows a typical plot of friction force vs pushing force in a simple pushing test. It can be seen that prior to its sliding motion, the frictional force will reach its maximum limit, which is denoted by F_m . This distinct feature can be used to quantify the surface roughness condition at the contact surface, which is known as the coefficient of static friction, μ_s . At this specific condition, the friction force is given by

$$F_m = \mu_s N \Rightarrow \mu_s = \frac{F_m}{N} \quad (2)$$

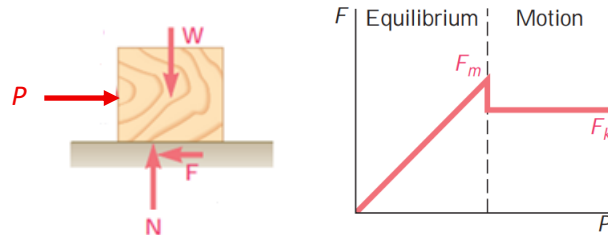


Fig. 3. Friction force in simple pushing test.

An alternate way to estimate the coefficient of static friction is to conduct the inclination test, as describe in section 4.2). It involves placing a block on an inclined platform, at which the angle of inclination is increased incrementally until the block starts to slide. Figure 4 shows the free-body diagram of a block resting on an inclined plane at angle θ .

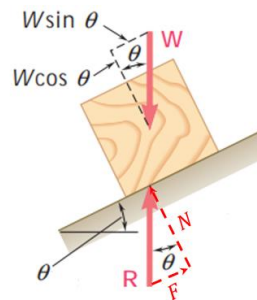


Fig. 4. Free-body diagram of a block on an inclined plane.

Resolving the weight W and contact force R into their components with respect to the inclined plane, their force equilibrium equations are:

$$N = W\cos\theta, \quad \text{and} \quad F = W\sin\theta \quad (3)$$

where the ratio of the two forces is given by

$$\frac{F}{N} = \frac{W\sin\theta}{W\cos\theta} = \tan\theta \quad (4)$$

Now, if the inclination reaches an angle Φ at which the block starts to slide downwards, it is an indication that the friction force has just passed its maximum friction force of F_m . Substitute this value into Eq. (4) and based on Eq. (2), the coefficient of static friction can be estimated to be

$$\mu_s = \tan\Phi \quad (5)$$

3) OBJECTIVE

In this lab session, we will investigate the use of Statics in two experiments, namely:

- i) *Statics of mystery bar* – Applying static analysis to determine the compositions of a heterogeneous bar that is made up of aluminium and steel parts.
- ii) *Static Friction* – Using the inclination test to investigate the characteristics of the coefficient of static friction.

4) EXPERIMENT SETUP AND PROCEDURES

4.1) Statics of mystery bar

You are provided with a heterogeneous bar with a steel part sandwiched between the aluminium parts, as shown in Fig. 5. You are required to determine the lengths (L_1 , L_2 and L_3) of each part using only digital balance provided.

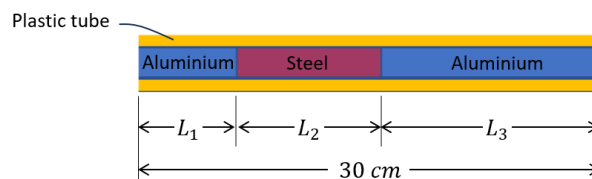


Fig. 5. Mystery heterogeneous bar.

The experimental procedures are as follows:

- Determine the *mass per unit length* of the steel, aluminum and plastic tube with the samples provided. They are denoted by m_s , m_a and m_p , respectively.
- Place the heterogeneous bar with its two ends on the balance machines and record the corresponding reaction forces R_1 and R_2 at these supports.
- Draw the free-body diagram of the above simply supported bar, and derive the static equilibrium equations, namely the vertical force equation and a moment equation about the left-end support. Note that for each metal part and the plastic tube, their effective weights are acting at their centroids, respectively.
- Together with the total length equation, i.e. $L_1 + L_2 + L_3 = 300 \text{ mm}$, we have exactly three equations to solve for the three unknown lengths uniquely. However, the moment equation is quadratic, and this makes the solution very challenging.

- Instead, let us try to explore using the principle of superposition to solve this problem. First, it should be noted that the mystery bar problem can also be represented by the sum of three separate cases, as shown in Fig. 6. In this case, the middle part is replaced by a fictitious material with density $m_s - m_a$, while the aluminum bar is made full-length. By doing so, the reaction forces for the full aluminum bar and plastic tube cases are straight-forward, which are half the weights at the two ends.

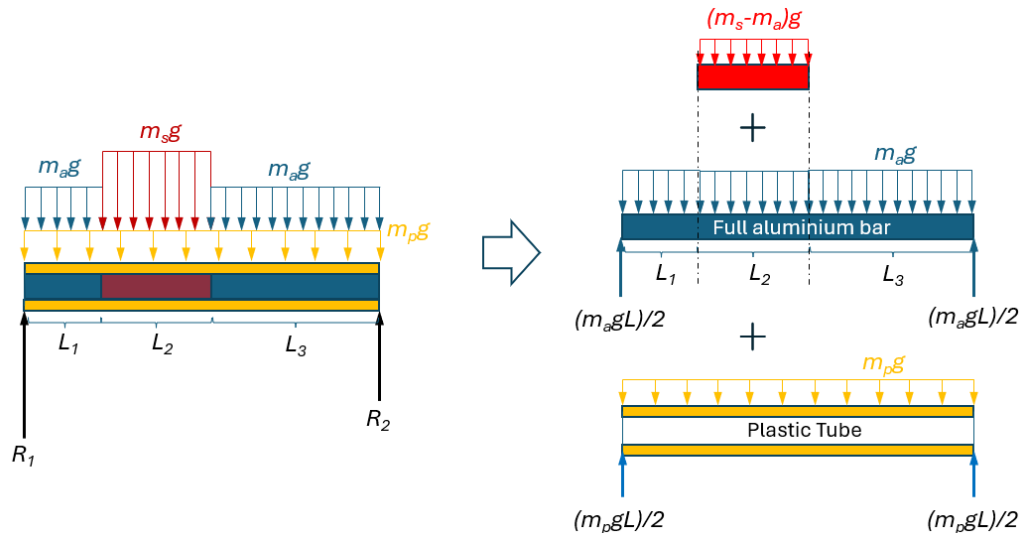


Fig. 6. Equivalent of the mystery bar with the superposition of three separate uniform bars.

Now, by subtracting loadings of the full aluminum bar and plastic tube from the original problem, we arrive at a fictitious problem that shows the middle bar is supported by a weightless tube at the two ends. Finally, the three lengths can be solved with the force and moment equations from this problem as depicted in Fig. 7.

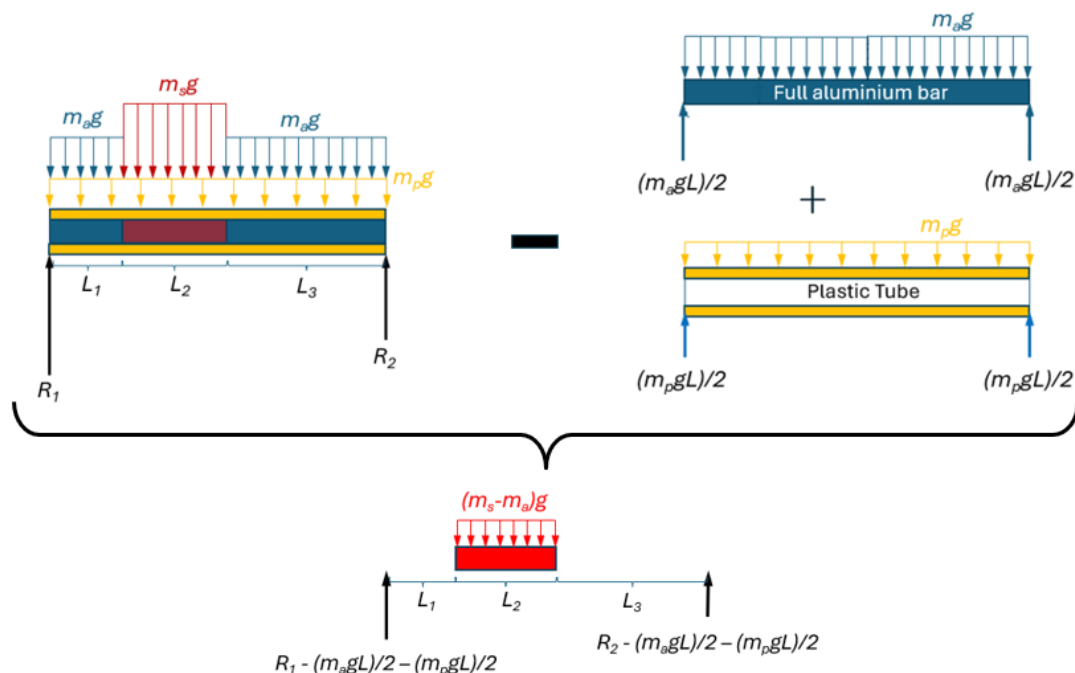


Fig. 7. Fictitious bar which is equivalent to the original mystery bar problem.

- Once you have determined the answers from your measurements and calculations, uncover the bar to measure the actual lengths of the three metal pieces. Compare them with your calculated values and explain their discrepancy and how would you improve the accuracy of the experiment.

4.2) Static Friction

In this experiment, we will be investigating the properties of static friction using the inclination test. As discussed in section 2.3, the coefficient of static friction can be estimated from the inclination angle at which the block starts to slide downward.

The experiment is conducted on the [vertical automatic handy test-stand](#), as shown in Fig. 8. The system is made up of a handy digital force gauge that is mounted on an automatic vertical servo control test stand. From the schematic diagram in Fig. 9, the inclination angle is given by

$$\phi = \sin^{-1}\left(\frac{H}{L}\right) \quad (6)$$

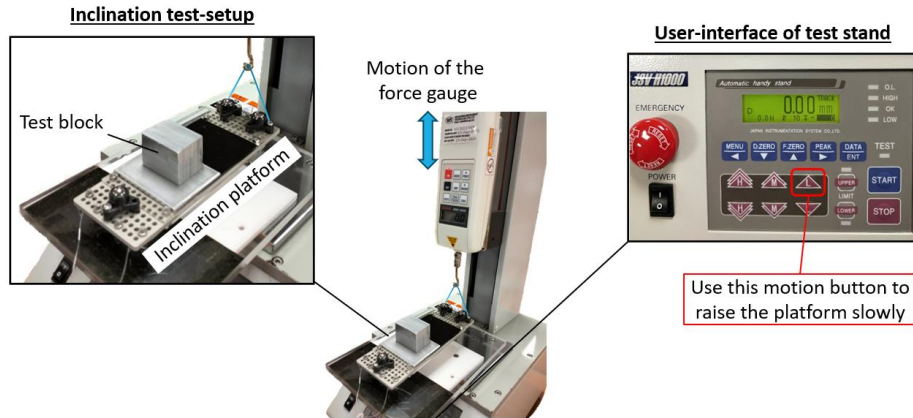


Fig. 8. Experimental setup of the vertical automatic handy test-stand.

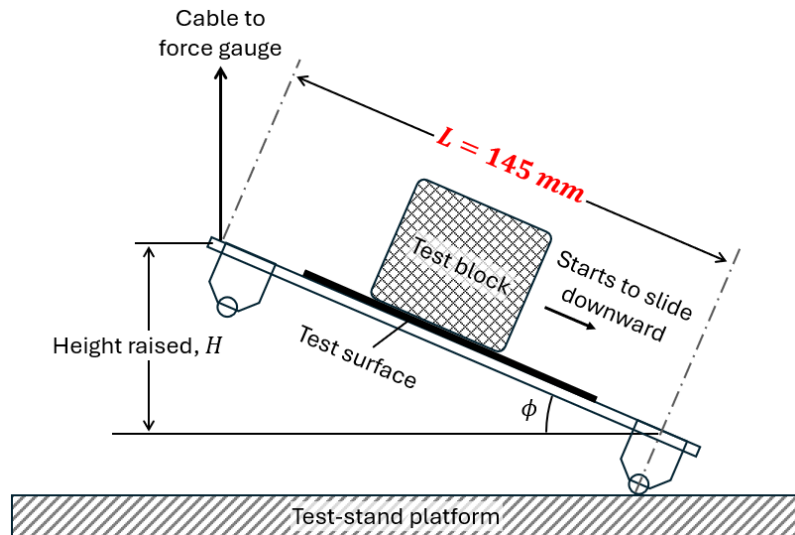


Fig. 9. Schematic diagram of the experiment.

The procedures for this experiment are as follows:

- Establish the reference position of the setup at which the test surface is at zero inclination angle. With the cable on the hook of the force gauge, press on the **L** motion button to raise it slowly, until the force gauge registers a significant force ($\approx 0.5\text{ N}$). At this position, you should observe that the cable is taut, and the test surface remains flat. Press the **D.ZERO** button to set this initial position.

- For all the tests in the following three sets of experiments, you are required to perform the following steps:
 - First, make sure that test-surface is at zero-inclination position, i.e. the displacement display is 0.00 mm.
 - Place the appropriate block on the center of test-surface.
 - Press the **START** button to move the force gauge upward continuously.
 - When you observe the block starts to slide downwards, press on the **STOP** button to cease the motion immediately.
 - Record the displacement, i.e. the height raised, H .
- Study the effect of mass on the static friction coefficient. In this experiment, you will use a single aluminum block and a double aluminum block (glue together) on the aluminum test-surface. Make sure that the contact surface areas of the blocks (length x breadth) are identical in both tests.
- Study the effect of contact surface area on the static friction coefficient. In this experiment, you will use the small aluminum block on the aluminum test-surface. In this case, make sure the smaller surface (length x height) of the block is in contact with the test-surface.
- Tabulate the static friction coefficients for different combinations of contact interfaces. In this experiment, you will test four different combinations with two test-surfaces (aluminum and rubber) and two block surfaces (aluminum and acrylic).
- With the inclination angles, calculate the static friction coefficients for the different contact interfaces.

Finally, complete the log sheet provided and submit the **GROUP** report onto the lab folder in CANVAS by the end of the lab session.